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Research Topic: Economic Trade-offs of Restricting Development in Flood-Prone Areas: A Case Study of Melamchi Municipality, Nepal

Abstract

This research will investigate the economic trade-offs of restricting development in flood-prone areas and future flood losses avoided in Melamchi Municipality, Nepal. The study will compare two development scenarios: continued development in flood-prone areas and restrictions on new development in high-risk flood-prone areas. A quantitative case study approach will be used, combining secondary data, GIS-based spatial analysis, economic valuation and scenario analysis. The opportunity cost of development restrictions and future flood losses and potential losses avoided will be estimated. Sensitivity analysis will examine how changes in key assumptions affect the estimated trade-off. The study is expected to provide a quantitative assessment of the economic implications of restricting development in flood-prone areas and contribute evidence that can support future land-use planning and flood-risk management decisions in Melamchi Municipality.

Introduction

Nepal is a landlocked country in South Asia, located in the Himalayan region [2]. The country lies between the Indian and Eurasian tectonic plates, which makes it a tectonically active region [1]. Its geographically complicated setting, combined with heavy monsoon rain, rugged topography and human activities has caused it to face recurrent floods and other water-induced disasters [2], [3].

Melamchi Municipality provides a relevant example of these flood-related risks. On 15 June, 2021, Melamchi faced a devastating flood that caused extensive damage to settlements, businesses, agricultural land and livelihoods [3]. The flood resulted in the destruction of 259 enterprises, loss of 3,500 ropani of khet land, and damage to roads and bridges, disrupting transportation and economic activities in affected areas [3]. These impacts demonstrate the economic consequences that flooding can have on communities in Melamchi Municipality [3].

These economic costs create a reason to consider limiting development in flood-prone areas. By limiting development in these areas, economic exposure and vulnerability to flood events as well as future flood damages can be reduced [6], [7], [8]. However, preventing the use of flood-prone land can reduce economic and livelihood opportunities such as agriculture and fishing [5]. This represents an opportunity cost, as restrictions on development may result in the loss of potential economic benefits from alternative land uses [6], [7]. Thus, there is an economic trade-off between limiting development in flood-prone areas and the potential economic costs of future flooding [6].

Existing Literature

Previous research has examined the economic costs and benefits associated with limiting development in flood-prone areas. Juarez-Lucas et al. [5] demonstrated, using a case study in Candaba, Philippines, the importance of considering the economic benefits from using flood-prone land for activities such as fishing and agriculture alongside flood risk. Hudson and Botzen [6] reviewed existing cost-benefit approaches to flood-zoning policies and identified cost-benefit analysis as a method of evaluating the consequences of limiting development in areas prone to floods. Kousky and Walls [7] used this method by examining floodplain conservation as a flood-mitigation strategy. They calculated the opportunity cost of keeping floodplain land out of development and the flood damages avoided because the land was protected. Johnson et al. [8] further examined this by studying the long-term economic effects of preventing floodplain development by comparing land-acquisition costs with future avoided flood damages.

Research has also examined how land-use planning and future development affect future flood risk and its economic consequences. Prose [9] developed a risk-based zoning approach for urbanising floodplains, considering the economic benefits of development alongside the additional risk created. Hamers et al. [11] developed a framework that considered changes in climate, socioeconomic development, and land-use planning for assessing future flood risk. Their results highlighted how socioeconomic development had a greater influence on future flood risk than climate change, increasing runoff and the value of economic assets exposed to flooding.

There has also been research on the economic impacts of flood exposure and the relationship between development and flood risk in Nepal. Delalay et al. [4] investigated future flood damage in Sindhupalchok, Nepal by assessing direct and indirect damage to hydropower plants, roads, houses, and farmland under different exposure scenarios. Banstola and Aldrich [12] examined how rapid urban growth and expansion is connected to changes in flood hazards in Kathmandu Metropolitan City. Mesta et al. [13] modelled future urban growth in Kathmandu Valley, Nepal to 2050 using the Sleuth model and found that built-up areas within the 100-year floodplain are projected to increase from 38 km² to 83 km², indicating that continued urban expansion could increase development exposed to flooding.

Existing research has examined the economic costs and benefits of limiting development in flood-prone areas and how land-use planning can reduce or increase future flood risks internationally [5]-[11], while studies in Nepal have focused on flood damage, urban growth and exposure to flood hazards [4], [12], [13]. However, there is limited research that directly compares the economic benefits of restricting development in flood-prone areas with the economic opportunities that may be lost due to these restrictions in Melamchi Municipality, Nepal. This study aims to address this gap by evaluating the economic trade-off of limiting development in flood-prone areas.

Research Question

Central Research Question:

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1. What is the economic trade-off between the opportunity costs of restricting development in flood-prone areas and the expected future flood losses avoided in Melamchi Municipality, Nepal?

Sub-Research Questions:

1. What are the potential economic opportunity costs of restricting development in flood-prone areas of Melamchi Municipality?
2. What are the expected economic losses from future flooding in these areas?
3. How do the opportunity costs of restricting development compare with the expected flood losses avoided?

Methodology

Research Design

This research will use a quantitative method and a case-study approach to investigate the economic trade-off between restricting development in flood-prone areas and expected future flood losses avoided in Melamchi Municipality. The research will primarily use secondary data, GIS-based spatial analysis, economic valuation and scenario analysis.

Data Collection

Secondary data will be collected from government agencies, research institutions, published studies, and publicly available datasets. Data will include historical flood events, associated economic damages, flood-prone areas, land-use patterns, agricultural activity, businesses, infrastructure and other economic assets within Melamchi Municipality. Existing research on the 2021 Melamchi flood will be used to study historical economic impacts. Data from multiple sources will be cross-checked where possible to improve reliability.

Spatial Analysis

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GIS will be used to identify flood-prone areas within Melamchi Municipality and determine the land uses, economic activities and developments located within these areas. Flood-hazard maps will be used alongside land-use and development data to identify economic assets exposed to flooding and areas that could be subjected to development restrictions to reduce future economic exposure.

Scenario Development and Variables

Two main scenarios will be developed.

Scenario 1 — Continued Development:
This scenario will represent continued development within flood-prone areas according to existing land-use patterns. The economic assets and activities exposed to flooding will be identified, and their potential future flood losses will be estimated.

Scenario 2 — Development Restriction:
This scenario will represent restrictions on new development within identified high-risk flood-prone areas. The restriction will focus on development opportunities that would otherwise occur without the restriction. The economic value of the development opportunities forgone will be estimated, together with the economic exposure and expected future flood losses.

Therefore, the main independent variable will be the development scenario, while the key outcome variables will be the opportunity cost of development restriction, expected future flood losses and flood losses avoided.

Economic Valuation

The opportunity cost of restricting development will be estimated through the economic value of activities and development opportunities forgone under Scenario 2. Depending on data availability, this may include agricultural production, existing businesses and potential land development. Where sufficient data are available, the value of each economic category will be estimated using an appropriate monetary measure. The assumptions and valuation methods used will be clearly stated and justified.

Estimation of Future Flood Losses

Expected future flood losses under Scenario 1 will be estimated using historical flood-damage information and economic assets exposed to future flooding within flood-prone areas. Information from the 2021 Melamchi flood and other available historical flood events will be used to demonstrate the relationship between flood events and economic damage. Potential losses to infrastructure, agriculture, businesses, and other measurable economic assets will be considered. Where sufficient data are available, different flood risk and severity levels will be incorporated into the estimates. The same types of economic assets will be considered under Scenario 2 to estimate future flood losses and potential losses avoided through development restrictions.

Trade-off and Data Analysis

The expected economic outcomes under the two scenarios will be compared to assess the trade-off between development restriction and flood risk. The opportunity cost of restricting development under Scenario 2 will be compared against the expected future

flood losses under Scenario 1. The difference in expected flood losses will then be used to estimate potential flood losses avoided by implementing development restrictions. Economic and flood-damage data will be summarised using descriptive statistics, with results expressed in monetary terms when reliable valuation is possible. Sensitivity analysis will vary key assumptions to determine how uncertainty in economic values, flood-loss estimates and flood severity affects the results.

Reliability, Ethics and Risk Management

Data from multiple credible sources will be cross-checked where possible to improve reliability. All sources will be appropriately cited, and any personal or confidential information obtained during the research will be protected. Potential risks such as incomplete local data, inconsistent datasets and difficulties accessing information will be addressed through alternative credible sources and transparent documentation of assumptions.

Expected Outcomes

This research is expected to produce estimates of the opportunity costs associated with restricting development in flood-prone areas and the expected future flood losses that could be avoided. By comparing the two scenarios, a quantitative assessment of the economic trade-off of restricting development in flood-prone areas of Melamchi Municipality is expected. The study may also identify which economic activity or assets contributes most significantly to this economic trade-off.

Limitations

Potential limitations include the lack of local economic data, uncertainty when estimating future flood losses, assumptions required to assign monetary values to development opportunities and the difficulty of representing all economic and social consequences in monetary terms. As a case study, the findings will primarily apply to Melamchi Municipality, Nepal and should not be generalised to all other areas of Nepal.

Conclusion

Flood-prone areas present a difficult land-use decision for Melamchi Municipality as restricting development may reduce future flood loss but would also limit potential economic opportunities. This creates an economic trade-off that should be taken into consideration when making decisions about development in areas exposed to flooding. By examining both the economic cost of restricting development and the potential costs avoided because of this restriction, this research aims to provide a clearer understanding of this economic trade-off.

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